

Modbus RTU Register Map – Version Gt03.0

TRIO-50.0/60.0-TL-OUTD

TRIO-TM-50.0-400

TRIO-TM-60.0-480



General liability warnings concerning inverter use

Please refer to the TRIO-50/60.0-TL-OUTD and TRIO-TM-50.0/60.0-400/480 Product Manuals for complete installation instructions and product use.



ABB accepts no liability for failure to comply with the instructions for correct installation and will not be held responsible for systems upstream or downstream the equipment it has supplied. It is absolutely forbidden to modify the equipment. Any modification, manipulation, or alteration not expressly agreed with the manufacturer, concerning either hardware or software, shall result in the immediate cancellation of the warranty.

The Customer is fully liable for any modifications made to the system.

Given the countless array of system configurations and installation environments possible, it is essential to check the following: sufficient space suitable for housing the equipment; airborne noise produced depending on the environment; potential flammability hazards.

ABB will NOT be held liable for defects or malfunctions arising from: improper use of the equipment; deterioration resulting from transportation or particular environmental conditions; performing maintenance incorrectly or not at all; tampering or unsafe repairs; use or installation by unqualified personnel.

ABB will NOT be held responsible for the disposal of: displays, cables, batteries, accumulators etc. The Customer shall therefore arrange for the disposal of substances potentially harmful for the environment in accordance with the legislation in force within the country of installation.

Field of use, general conditions

ABB shall not be liable for any damages whatsoever that may result from incorrect or careless operations.



You may not use the equipment for a use that does not conform to that provided for in the field of use. The equipment MUST NOT be used by inexperienced staff, or even experienced staff if carrying out operations on the equipment that fail to comply with the indications in this manual and enclosed documentation.

Intended or allowed use

This equipment is a multi-string inverter designed for:

- Transforming a continuous electrical current (DC)
- supplied by a photovoltaic generator (FV)
- in an alternating electrical current (AC)
- suitable for feeding into the public distribution network

P/N TRIO-50.0/60.0-TL-OUTD
P/N TRIO-TM-50.0-400
P/N TRIO-TM-60.0-480

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Introduction

The purpose of this document is to describe the Modbus RTU registers map for the monitoring and control of the inverter TRIO-50.0/60.0-TL-OUTD and TRIO-50.0/60.0-TM-OUTD inverter families by an external modbus RTU master over a RS-485 serial line.

Modbus RTU is a Master-Slave communication protocol over serial line with defined frames, nomenclature, physical layer, Cyclic Redundancy Code. The inverter TRIO-50.0/60.0-TL is compatible with the standard and applies to the protocol features described on this document, any other feature of the protocol not described on this document is not supported.

The inverter publishes two different and separate set of registers that contain different set of parameters and options: *Holding Register* and *Input Register*.

- *Holding Register* are Read (Function 3) and Write (Function 6 and 16) registers used mainly for controlling purpose. Any command sent to the inverter must be addressed through Holding registers.
- *Input Register* are Read Only (Function 4) registers used to gather status and measures from the inverter.

Modbus addressing model and glossary

The inverter Modbus map refers to the Modbus Data Model, therefore an Offset between Modbus Data Address and Modbus PDU must be taken into consideration as described on the Modbus Protocol:

“The MODBUS application protocol defines precisely PDU addressing rules. In a MODBUS PDU each data is addressed from 0 to 65535. It also defines clearly a MODBUS data model composed of 4 blocks that comprises several elements numbered from 1 to n. In the MODBUS data Model each element within a data block is numbered from 1 to n. Afterwards the MODBUS data model has to be bound to the device application”



MODBUS DATA numbered X is addressed in the MODBUS PDU X - 1.

Please refer to the following definitions for details about these keywords:

Modbus Request: The data sharing on Modbus. Each request on Modbus starts from the Master (unique for each communication).

Slave ID: The serial line RS485 address that defines each device in the communication bus, each device must have a unique Slave ID different from all the other devices connected to the same serial line. According to the standard the Slave ID must have a unique address from 1 to 247 while the address 0 is reserved for Broadcast requests.

Modbus Frame: The sequence of bit transmitted on a Modbus Request. Each Frame must be separated by the next one by a guard band of at least 3.5 characters.

Modbus Register (PDU Register): The register written on the payload of a Modbus Frame. The Modbus Registers range is between 0 and 65535.

Modbus Data Address: The address assigned to a Modbus Register according to the Data Address Nomenclature. The Modbus Data Address range is between 1 and 65536 and can be translated as Modbus Register + 1.

Length: The field of the Modbus Frame that includes the number of registers to be considered on the Modbus Request.

Function: The Modbus function as described on the Modbus protocol.

Exception Code: The code received in case of communication failure. The Modbus protocol defines a list of exception codes used to describe an unwanted behaviour that may happen on a Modbus Request. For example a request to a not implemented function can trigger an exception code 1 Illegal Function.

Serial line configuration

The inverter serial Line can be configured as follows:

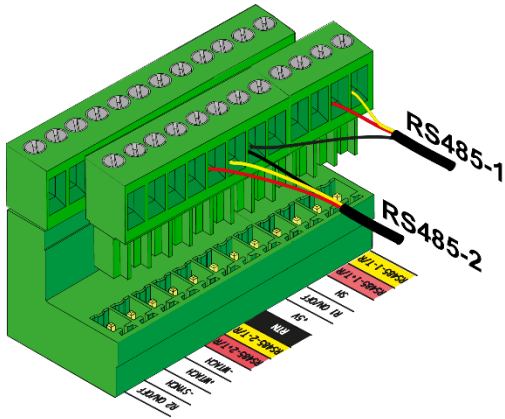


Figure 1: Inverter Serial Ports

Serial line RS 485-1	Serial line RS 485-2
Aurora Protocol	Aurora Protocol
Modbus RTU Slave*	Modbus RTU Slave*
	Modbus Sunspec

* This document refers only to the proprietary Modbus map identified as “Modbus RTU Slave”. For details about SunSpec protocol and “SunSpec” Modbus map please refer to the Sunspec Alliance website: <http://sunspec.org>.

For further information about the RS485 serial line connection please refer to the inverter’s product manual available on ABB official website www.abb.com/solarinverters

The configuration of the serial line (protocol selection, Slave ID, Baud rate, parity and bit stop) must be done through the software “Aurora Manager Lite”. For further information about the software “Aurora Manager Lite” please refer to the software manual available on the ABB official website www.abb.com/solarinverters

Peripheral settings

- **Interface:** RS-485 (half duplex)
- **Baud Rate:** 2400, 4800, 9600, 19200 (default value), 38400, 57600 or 115200bps
- **Start bit:** 1
- **Stop bit:** 1
- **Parity:** No parity (default value), even parity or odd parity
- **Data bits:** 8
- **Byte order:** Big-endian
- **Bit order:** Less Significant Bit (LSB) sent first
- **Minimum Timeout:**100ms

Modbus Register Map version

- **Version:** Gt03
- **Minimum inverter update version:** “1711C” for Europe version; “1711D” for USA version, “1708B” for Japan version

Modbus Register Map version history table

Modbus RTU Register Map version	Inverter Update Version (First version released)							
	TRIO-XX-YY-OUTD (EU)				TRIO-XX-YY-OUTD (US)			TRIO-50-TL-OUTD-JP
	XX=50 YY=TL	XX=TM YY=50	XX=60 YY=TL	XX=TM YY=60	XX=50 YY=TL	XX=60 YY=TL	XX=TM YY=60	
GU0.0	1639D	-	1639E	-	-	-	-	-
Gt 3.0	1711C	1741E	1715B	1741F	1711D	1715C	-	1708B

Function codes supported

- 03 (0x03) Read holding registers;
- 04 (0x04) Read input registers;
- 06 (0x06) Write single register;
- 16 (0x10) Write multiple registers.



Note: a Modbus request with the functions not included on the above compatibility list will trigger an exception code 01, Illegal function.

Supported exception codes

- 01 Illegal function
- 02 Illegal data address
- 03 Illegal data value
- 04 Server device failure
- 06 Server device busy



NOTE: for every modbus request that includes a register not mapped, the inverter will trigger an exception code 02, illegal data address.



Note: Any Modbus request Modbus Data Address not included on this document will trigger an exception with code 2.

Modbus RTU frame

The inverter applies the same Modbus RTU frames defined on the protocol. Each Modbus data handshake starts with a Modbus Request from the master (unique according to the standard). The Master request frame has the following structure:

Functions #3 and Function #4

Slave ID	Function	Start PDU Register	Length	CRC
1 byte	1 byte	2 bytes	2 bytes	2 bytes

Function #6

Slave ID	Function	PDU Register	Register Value	CRC
1 byte	1 byte	2 bytes	2 bytes	2 bytes

Function 16

Slave ID	Function	Start PDU Register	Number of Registers	Byte Count	Data Values	CRC
1 byte	1 byte	2 bytes	2 bytes	1 byte	Byte Count	2 bytes

Slave answer PDU for Functions #3 and Function #4

Slave ID	Function	Byte Count	Data (Registers Value)	CRC
1 byte	1 byte	1 byte	Byte Count	2 bytes

Slave answer PDU for Functions #6

Slave ID	Function	PDU Register	Register Value	CRC
1 byte	1 byte	2 bytes	2 bytes	2 bytes

Slave answer PDU for Functions #16

Slave ID	Function	Start PDU Register	Number of Registers	CRC
1 byte	1 byte	2 bytes	2 bytes	2 bytes

The maximum size of any MODBUS RTU frame is 256 bytes. No Slave Answer is returned for Broadcast commands.

Data Encoding

The Data types available on this document refer to:

- Unsigned Integer, 16-bit Modbus registers (U16)
- Unsigned Integer, 32-bit Modbus registers (U32)
- Signed Integer, 16-bit Modbus registers (S16)
- Word Swapped Floating Point 32-bit, 32 bit Modbus registers (SF32)
- Ascii character encoded on Uint 16 (ASCII string)

The cyclic redundancy code is Modbus CRC16, a coding example of CRC can be found on annex 13

Unsigned integer 16-bit

Unsigned integer data is available as a 16 bit value encoded on a single Modbus PDU register.

Data encoded as U16	
U16 MSB	U16 LSB

Modbus PDU register	
Register MSB	Register LSB

An Unsigned integer value 11 translates as follows:

Value	Data encoded as UInt16		Modbus PDU register	
11	U16 MSB	U16 LSB	Register MSB	Register LSB
	0x00	0x0B	0	0x0B

Example: The following data exchange represents the request of global state to the inverter with address 2, the inverter answer is Run (Global State = 6 according to Annex 5)

Request (Hexadecimal format):

Slave ID	Function	Start PDU Register		Length		CRC	
0x02	0x04	0x04	0x19	0x00	0x01	0xE1	0x0E

Answer (hexadecimal format):

Slave ID	Function	Byte Count	Data (Registers Value)		CRC	
0x02	0x04	0x02	0x00	0x06	0x7D	0x32



Note: The offset between the Modbus PDU Register (0x419 = 1049) and the Modbus Data Address (1050).

Unsigned integer 32-bit

Unsigned integer data is available as a 32 bit value encoded on two Modbus PDU register.

Data encoded as U32				Modbus PDU register			
U32 High MSB	U32 High LSB	U32 Low MSB	U32 Low LSB	Register High MSB	Register High LSB	Register Low MSB	Register Low LSB

An unsigned integer value 323000 translates as follows:

Value	Data encoded as U32				Modbus PDU register			
323000	U32 High MSB	U32 High LSB	U32 Low MSB	U32 Low LSB	Register High MSB	Register High LSB	Register Low MSB	Register Low LSB
	0x00	0x04	0xED	0xB8	0x00	0x04	0xED	0xB8

Example: The following data exchange represents the request of total energy to the inverter with address 2, the inverter answer is 153000 kWh.

Request (hexadecimal format):

Slave ID	Function	Start PDU Register		Length		CRC	
0x02	0x04	0x04	0x2F	0x00	0x02	0x41	0x01

Answer (Hexadecimal format):

Slave ID	Function	Byte Count	Data (Registers Value)				CRC	
0x02	0x04	0x04	0x00	0x02	0x55	0xA8	0x57	0xAA



Note: The offset between the Modbus PDU Register (0x42F = 1071) and the Modbus Data Address (1072).

Signed integer 16-bit

Signed integer data is available as a 16 bit value encoded on a single Modbus PDU register

Data encoded as 16		Modbus PDU register	
S16 MSB	S16 LSB	Register MSB	Register LSB

An unsigned integer value 11 translates as follows:

Value	Data encoded as S16		Modbus PDU register	
-22	U16 MSB	U16 LSB	Register MSB	Register LSB
	FF	EA	FF	EA

Example: The following data exchange represents the Modbus request for a new reactive power set point (Modbus Data Address 507 on holding register) equal to -100% of nameplate rating (over-excited set point) for all the inverter connected to the communication bus with a Broadcast command.

Request (Hexadecimal format):

Slave ID	Function	Start PDU Register		Number of Registers		Byte Count	Data Values		CRC	
0x00	0x10	0x01	0xFA	0x00	0x01	0x02	0xFF	0x9C	0xEE	0xC3

No Answer is received when the Modbus Request is Broadcast.



Note the offset between the Modbus PDU Register (0x1FA = 506) and the Modbus Data Address (507).

Word Swapped Floating Point 32-bit (SF32)

Word Swapped Floating Point is data type for IEEE754 Floating point where the two Modbus Registers are swapped before processing the floating point. Word Swapped Floating Point data is available as a 32 bit value encoded inside two Modbus PDU registers.

Data encoded as SF32				Modbus PDU register			
SF32 High MSB	SF32 High LSB	SF32 Low MSB	SF32 Low LSB	Register Low MSB	Register Low LSB	Register High MSB	Register High LSB

A Word Swapped Floating Point Value 25.52 translates as follows:

Value	Data encoded as U32				Modbus PDU register			
25.52	U32 High MSB	U32 High LSB	U32 Low MSB	U32 Low LSB	Register Low MSB	Register Low LSB	Register High MSB	Register High LSB
	0x41	0xCC	0x28	0xF6	0x28	0xF6	0x41	0xCC



Note the Modbus Registers are swapped before processing the IEEE754 Floating Point.

Example: the following data exchange represents the request of output power to the inverter with address 2, the inverter answer is 20000W.

Request (Hexadecimal format):

Slave ID	Function	Start PDU Register		Length		CRC	
0x02	0x04	0x04	0x45	0x00	0x02	0x61	0x1D

Answer (Hexadecimal format):

Slave ID	Function	Byte Count	Data (Registers Value)			CRC	
0x02	0x04	0x04	0x40	0x00	0x46	0x9C	0xEF



Note the offset between the Modbus PDU Register (0x445 = 1093) and the Modbus Data Address (1094).

Ascii character 16-bit

Ascii character are encoded as Unsigned Int16 data type but the value is processed as Ascii Code.

Data encoded ASCII String		Modbus PDU register	
ASCII String MSB	ASCII String LSB	Register MSB	Register LSB

An Ascii character “1” translates as follows:

Value	Data encoded as U16		Modbus PDU register	
“1”	U16 MSB	U16 LSB	Register MSB	Register LSB
	0x00	0x31	0x00	0x31

Example: the following data exchange represents the request of serial number to the inverter with address 2, the inverter answer is 123789

Request (Hexadecimal format):

Slave ID	Function	Start PDU Register		Length		CRC	
0x02	0x04	0x03	0xF0	0x00	0x06	0x70	0x4C

Answer (Hexadecimal format):

Slave ID	Function	Byte Count	Data (Registers Value)						CRC	
0x02	0x04	0x0C	0x0031	0x0032	0x0033	0x0037	0x0038	0x0039	0xD7	0x33



Note the offset between the Modbus PDU Register (0x3F0 = 1008) and the Modbus Data Address (1009).

Capability

The capability, shown on Fig.2, defines the limits for active and reactive power of the inverter output. The capability limit reduces according to the voltage level at the inverter terminal output with a rate of 1,06% P/P_N each 1% voltage reduction, starting from 93,75 V_N at inverter terminal output.

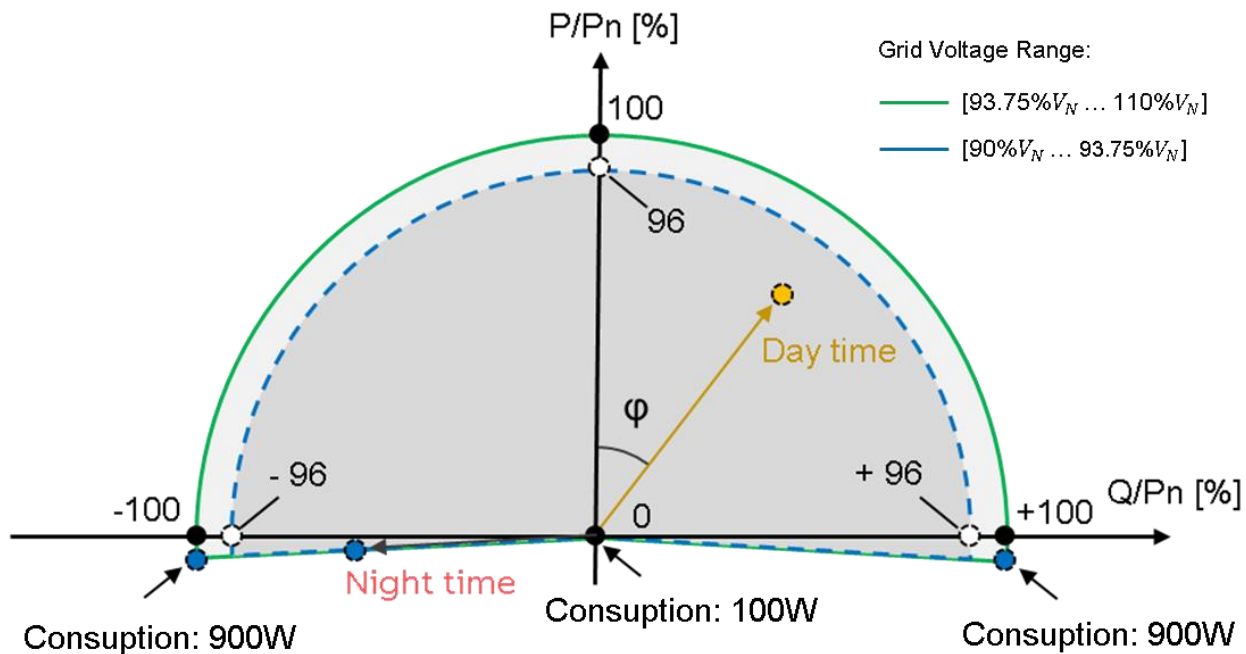


Figure 2: Inverter Capability

Conditions	Ratings				
	TRIO-50-EU	TRIO-60-EU	TRIO-50-US	TRIO-60-US	TRIO-50-JP
Nominal apparent power [S_n]	50 kVA	60 kVA	50 kVA	60 kVA	50 kVA
Nominal Reactive Power [Q_n]	50 kVAr	60 kVAr	50 kVAr	60 kVAr	50 kVAr
Nominal Active Power [P_n]	50 kW	60 kW	50 kW	60 kW	50 kW
Nominal Voltage [V_n]	400 Vac	480 Vac	480 Vac	480 Vac	420/440 Vac
Ambiente Temperature	$\leq 50^\circ\text{C}$	$\leq 45^\circ\text{C}$	$\leq 50^\circ\text{C}$	$\leq 45^\circ\text{C}$	$\leq 50^\circ\text{C}$



NOTE: the inverter performs the set points with reactive power priority, if a set point of reactive power requires that the inverter works out of the capability profile, then the inverter will limit the active power to the maximum allowed by the capability profile.



NOTE: although the inverter can provide reactive power during night time, it is necessary to configure properly the inverter to be able to operate during night time

Power management set points

The power management commands are a set of dedicated registers allowing to change the inverter output power. The Modbus map includes two distinct areas for power management:

- **Block #1 backward compatibility block:** The modbus holding registers in the range [190..227] are a dedicated Modbus block for inverter backwards compatibility with legacy ABB Modbus conversion systems. It is strongly discouraged the use of the block #1 except for reasons of backward compatibility with converters PVI-RS485-Modbus.
- **Block #2 Plant controller block:** The modbus holding registers in the range [501..516] are a dedicated Modbus block for power management to be used in the development of centralized power control systems. The block #2 is also compatible with ABB native Modbus inverter, for example:
 - TRIO-20/27.6-TL-OUTD
 - ULTRA-700/1050/1400-TL-OUTD

The plant controller block allows the users to manage the power within a contiguous area and with higher accuracy on set points compared to the back compatibility block.



NOTE: The configuration of each block is not shared with the other blocks, so each setting realized on a block does not apply to the others.

Within each block it is possible to configure a set of parameters to manage the behavior of the inverter:

- **Control functions:** The inverter can receive a set point for Active Power Curtailment, Reactive Power and Power Factor. Reactive power and Power Factor control functions are mutually exclusive.
- **Dynamic and Permanent Commands:** Each set point can be sent with a Dynamic or Permanent mode. When using a dynamic control, the power set point will be active until the Timeout period has expired or another set point, within the same control function, updates the Timeout period. Once the Timeout period has elapsed, the inverter will come back to the default settings. When using a permanent control, the power set point will be always active. In addition, any configuration of the set point will be saved as new default settings (see next chapter).
- **Transient Options:** See annex 8
- **Transient Time:** See annex 9
- **Reset:** A reset command will force the inverter to a “no regulation” state.
- **Broadcast:** The broadcast commands are supported and are compatible with the specification of the Modbus protocol (Slave ID = 0).



NOTE: It is mandatory not to use permanent commands for power plant controllers: any permanent command will be written on inverter internal memory. Write continuously permanent set points can damage the internal memory of the inverter. For this purpose it is recommended to use permanent set points for default settings, safety or non-regulation situations and for power control with dynamic commands.

Before to send any set point to the inverter is recommended to complete the configuration of the above parameters, the available set points are: active power curtailment, reactive power and power factor (see Figure 3). Reactive power and power factor set points are mutually exclusive, whenever a new mode is selected through modbus data address 502, the previous mode will remain active until a set point is received by the inverter on the new mode.

The inverter shows always the values currently applied on set points, following a setup changes (smooth mode, timeout, transient time, Reg. 502 and 506), the inverter shows the last value configured properly. Set point and configurations return to their default values at the inverter power off.

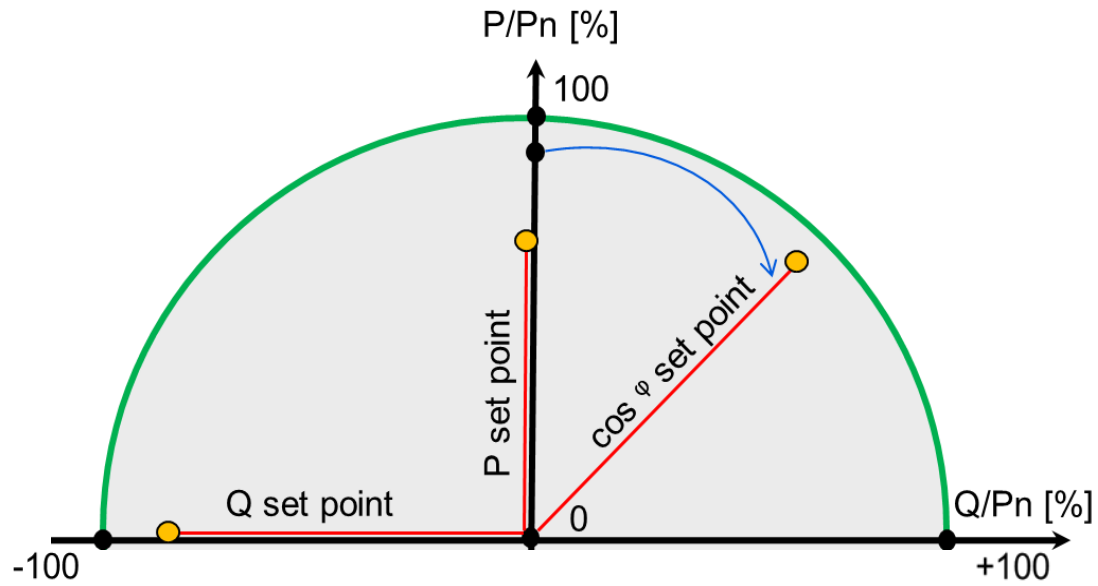


Figure 3: Available power control modes

Each set point received by the inverter will be elaborated by the internal logic board before being applied. Once the Modbus Request is received by the communication bus of the inverter, the time required to process a new set point may be estimated as follows.

Control Function	Command Type	Reaction Time
Active power curtailment	Dynamic or Permanent	≈ 40ms
Reactive Power	Dynamic or Permanent	≈ 10ms
Power Factor	Dynamic or Permanent	≈ 10ms

Registers map

Holding registers map

Modbus Data Address	Number of Registers	Register content description	Range	Default Value	Unit	Data Type	Notes
0180	1	Remote On/Off	0 or 1	0	-	U16	0 = Remote on 1 = Remote off This command will be executed only if the "Remote on/off" function is enabled in the inverter
0181	1	Remove "Reset by Hand" latch-state	0 or 1	0	-	U16	1 = Remove latch-state A request of reset will remove the latch state (if pending) then the register value will reset to 0. The latch-reset upon a grid fault is available by default only for Japanese grid code
Power management Block #1: Backward compatibility block							
0190	1	Transient Time for Permanent and Dynamic mode Holds time interval used by inverter when written commands are received to Modbus Data Addresses 0200, 0202, 0210, 0212, 0220, 0222, 0225 or 0227	-	4	s	-	See Annex 9
0191	1	Timeout for Dynamic mode Holds initial value of countdown timer used by inverter to time out a Dynamic Mode command. Timeout reset when new values are written on Data Address 0200, 0210, 0220 or 0225	0 to 250	2	min	U16	
0192	6	RESERVED					
0198	1	Transient Options for Permanent and Dynamic mode Smooth Mode and Transient Step for Modbus Data Addresses 0200, 0202, 0210, 0212, 0220, 0222, 0225 or 0227	0 or 128 or 256 or 384	256	-	-	See Annex 8
0199	1	RESERVED					

Modbus Data Address	Number of Registers	Register content description	Range	Default Value	Unit	Data Type	Notes
0200	2	Dynamic Mode Power Factor Set Point : Reactive Power expressed as fixed Power Factor	-1..-0.001 OR +0.001.. +1	0	-	SF32	Negative value for over-excited (capacitive) injection Positive value for under-excited (inductive) injection Must be used with Transient configurations (Data Address 0190, 0191,0198)
0202	2	Permanent Mode Power Factor Set Point : Reactive Power expressed as fixed Power Factor	-1..-0.001 OR +0.001.. +1	1	-	SF32	Negative value for over-excited (capacitive) injection Positive value for under-excited (inductive) injection Must be used with Transient configurations (Data Address 0190, 0191, 0198).
0204	1	Reserved					
0205	1	Reset Reactive Power or Power Factor (PF) management	0 or 1	0	-	U16	Setting the register to 1 will force the Reactive Power (PF) to reset from current value to zero (PF=1) Then the register value resets to 0
0206	4	Reserved					
0210	1	Dynamic Mode Active Power Set Point: Active Power Curtailment expressed as percentage of Nominal Power in % steps	0 to 100	100	%	U16	Must be used with Transient configurations (Data Address 0190, 0191, 0198)
0211	1	Reserved					
0212	1	Permanent Mode Active Power Set Point: Active Power Curtailment expressed as percentage of Nominal Power in % steps	0 to 100	100	%	U16	Must be used with Transient configurations (Data Address 0190, 0191, 0198)
0213	2	Reserved					
0215	1	Reset Active Power management	0 or 1	0	-	U16	Setting the register to 1 will force the active power to reset from current value to Nominal Power Then the register value resets to 0
0216	4	Reserved					

Modbus Data Address	Number of Registers	Register content description	Range	Default Value	Unit	Data Type	Notes
0220	2	Dynamic Mode Reactive Power Set Point: Reactive Power expressed as ratio of Max apparent power (in term of $\cos\phi$)	-1.0 to 1.0	0	-	SF32	Negative value for over-excited (capacitive) injection Positive value for under-excited (inductive) injection Must be used with Transient configurations (Data Address 0190, 0191, 0198)
0222	2	Permanent Mode Reactive Power Set Point: Reactive Power expressed as ratio of Max apparent power (in term of $\cos\phi$)	-1.0 to 1.0	0	-	SF32	Negative value for over-excited (capacitive) injection Positive value for under-excited (inductive) injection Must be used with Transient configurations (Data Address 0190, 0191, 0198)
0224	1	Reserved					
0225	1	Dynamic Mode Reactive Power Set Point: Reactive Power expressed as percentage of Max apparent power in % steps	-100 to 100	0	%	S16	Negative value for over-excited (capacitive) injection Positive value for under-excited (inductive) injection Must be used with Transient configurations (Data Address 0190, 0191, 0198)
0226	1	Reserved					
0227	1	Permanent Mode Reactive Power Set Point: Reactive Power expressed as percentage of Max apparent power in % steps	-100 to 100	0	%	S16	Negative value for over-excited (capacitive) injection Positive value for under-excited (inductive) injection Must be used with Transient configurations (Data Address 0190, 0191, 0198)
Heartbeat							
0300	1	Heartbeat counter	0 to 65535	0	s	U16	Increments every second The counter resets to 0 automatically when reaches 65535
Measures							
0301	2	Inverter - Grid Reactive Power	-	-	VAR	SF32	Read-only register If the measure of Reactive Power is not supported by the inverter, the register returns 0xFFFFFFFF

Modbus Data Address	Number of Registers	Register content description	Range	Default Value	Unit	Data Type	Notes
0301	2	Inverter - Grid Reactive Power	-	-	VAr	SF32	Read-only register If the measure of Reactive Power is not supported by the inverter, the register returns 0xFFFFFFFF
0303	2	Inverter - Grid Voltage	-	-	V	SF32	Read-only register
0305	2	Inverter - Grid Active Power	-	-	W	SF32	Read-only register
0307	2	Inverter - Grid Current	-	-	A	SF32	Read-only register
Power management Block #2: Plant controller block							
0501	1	Accuracy Set unit for Modbus Data Addresses 0507, 0508, 0511 and 0512 (% or ‰)	0 to 1	0	-	U16	0 = ‰ 1 = %
0502	1	Set active mode Permanent or Dynamic mode selection	0 to 1	0	.	U16	0 = Dynamic mode 1 = Permanent mode
0503	1	Transient Time for Permanent and Dynamic mode Holds time interval used by Inverter when are received write commands on Modbus Data Addresses 0507, 0508, 0509, 0511, 0512 or 0513	-	4	s	-	See Annex 9
0504	1	Timeout for Dynamic mode Holds initial value of countdown timer used by Inverter to time out a Dynamic Mode command. Timeout reset when new values are written on Data Address 0507, 0508 or 0509	0 to 250	2	min	U16	
0505	1	Transient Options for Permanent and Dynamic mode Smooth Mode and Transient Step for Modbus Data Addresses 0507, 0508, 0509, 0511, 0512 or 0513	0 or 128 or 256 or 384	256	-	-	See Annex 8
0506	1	Reactive Power control mode Set reactive power control mode on Reactive Power (Q fixed) or Power Factor (PF fixed)	0 to 1	1	-	U16	0 = PF fixed mode 1 = Q fixed mode

Modbus Data Address	Number of Registers	Register content description	Range	Default Value	Unit	Data Type	Notes
0507	1	Dynamic Mode Reactive Power Set Point: Reactive Power expressed as percentage of Nominal Power in % steps	100 to 100 or -1000 to 1000	0	% or ‰	S16	Negative value for over-excited (capacitive) injection Positive value for under-excited (inductive) injection Must be used with configurations on Data Address 0501, 0502, 0503, 0504, 0505 and 0506
0508	1	Dynamic Mode Active Power Set Point: Active Power Curtailment expressed as percentage of Nominal Power in % steps	0 to 100 or 0 to 1000	1000	‰	U16	Must be used with configurations on Data Address 0501, 0502, 0503, 0504 and 0505
0509	2	Dynamic Mode Power Factor Set Point : Reactive Power expressed as fixed Power Factor	-1.0 to 1.0	1	-	SF3 2	Negative value for over-excited (capacitive) injection Positive value for under-excited (inductive) injection Must be used with configurations on Data Address 0501, 0502, 0503, 0504, 0505 and 0506
0511	1	Permanent Mode Reactive Power Set Point: Reactive Power expressed as percentage of Nominal Power in % steps	100 to 100 or -1000 to 1000	0	% or ‰	S16	Negative value for over-excited (capacitive) injection Positive value for under-excited (inductive) injection Must be used with configurations on Data Address 0501, 0502, 0503, 0504, 0505 and 0506
0512	1	Permanent Mode Active Power Set Point: Active Power Curtailment expressed as percentage of Nominal Power in % steps	0 to 100 or 0 to 1000	1000	‰	U16	Must be used with configurations on Data Address 0501, 0502, 0503, 0504 and 0505
0513	2	Permanent Mode Power Factor Set Point : Reactive Power expressed as fixed Power Factor	-1.0 to 1.0	1	-	SF3 2	Negative value for over-excited (capacitive) injection Positive value for under-excited (inductive) injection Must be used with configurations on Data Address 0501, 0502, 0503, 0504, 0505 and 0506

Modbus Data Address	Number of Registers	Register content description	Range	Default Value	Unit	Data Type	Notes
0515	1	Reset Active Power management	0 or 1	0	-	U16	Setting the register to 1 will force the active power to reset from current value to Nominal Power Then the register value resets to 0
516	1	Reset Reactive Power or Power Factor (PF) management	0 or 1	0	-	U16	Setting the register to 1 will force the Reactive Power (PF) to reset from current value to zero (PF=1) Then the register value resets to 0
801	1	Enable/Disable “Reset by Hand” function	0 or 1	1	-	U16	0 = Disable “Reset by Hand” 1 = Enable “Reset by Hand”
802	1	Remove “Reset by Hand” latch-state	0 or 1	0	-	U16	1 = Remove latch-state A request of reset will remove the latch state (if pending) then the register value will reset to 0. The latch-reset upon a grid fault is available by default only for Japanese grid code Alternative to modbus data address 181(no difference)
Communication							
1006	1	Set communication protocol for serial line RS485#1	0 to 1	0	-	U16	0 = Aurora 1 = Modbus Setting any value other than 1 will trigger the change of the protocol immediately after the reply to the Modbus request
1007	1	Set communication protocol for serial line RS485#2	0 to 1	0	.-	U16	0 = Aurora 1 = Modbus Setting any value other than 1 will trigger the change of the protocol immediately after the reply to the Modbus request



Note: Configure properly the Modbus Address 0501 (Accuracy), 0502 (Mode), 0503 (Transient Time), 0504 (Timeout), 0505 (Transient Options) and 0506 (Reactive Power control mode) before to send any Set point



Note: Any reading Modbus request (Function #3) or writing Modbus request (Function #6 and Function #16) that start and end on available Modbus Data Address will not trigger any exception code, also if the request includes reserved register. Modbus request that start or end on reserved register will trigger an exception with code 2.



Note: Any Modbus request on 32 bit data type (SF32 and U32) must include both registers otherwise the inverter will trigger an exception with code 2.



Note: The convention used for the sign of the reactive power must be considered as the default value. Power Factor sign always follows the convention of the reactive power.

Input registers map

Modbus Data Address	Number of Registers	Register content description	Range	Unit	Data Type	Maximum Scan Rate (ms)	Notes
Inverter – Product Details							
1000	1	Inverter ID	261	-	U16	-	Fixed Value
1001	1	Inverter Block Modbus Map size	132	-	U16	-	Fixed Value
1002	1	Inverter Presence	0 or 1	-	U16	-	1 = Device present
1003	6	Inverter Part number	-	-	ASCII String	-	Fixed Value “-3N63-”
1009	6	Inverter Serial number	-	-	ASCII String	-	
1015	2	Inverter Manufacture date (Week)	-	-	ASCII String	-	
1017	2	Inverter Manufacture date (Year)	-	-	ASCII String	-	Registers value = Year of manufacture - 2000
1019	1	Inverter Type	-	-	ASCII String	-	See Annex 1
1020	1	Grid Type	-	-	ASCII String	-	See Annex 2
1021	1	Transformer Type	78	-	ASCII String	-	“N” = Transformerless
1022	1	Model Type	-	-	ASCII String	-	See annex 14
Inverter – States							
1050	1	Global State	-	-	U16	100	See Annex 3
1051	1	Alarm State	-	-	U16	100	See Annex 4
1052	1	DC/DC Converter State	-	-	U16	100	For Debug, see Annex 5
1053	1	DC/AC Converter State	-	-	U16	100	For Debug, see Annex 6
1054	1	Derating State	-	-	U16	100	See Annex 7
Inverter – Energy							
1070	2	Daily Energy	-	Wh	U32	100	
1072	2	Total Energy	-	Wh	U32	100	
1074	2	Partial Energy	-	Wh	U32	100	
1076	2	Weekly Energy	-	Wh	U32	100	
1078	2	Monthly Energy	-	Wh	U32	100	
1080	2	Yearly Energy	-	Wh	U32	100	
Inverter – Measures							
1090	2	Mean Grid Voltage (Phase to Neutral)	-	V	SF32	100	

Modbus Data Address	Number of Registers	Register content description	Range	Unit	Data Type	Maximum Scan Rate (ms)	Notes
1092	2	Mean Output Current (Phase to Neutral)	-	A	SF32	100	
1094	2	Output Active Power: Instantaneous	-	W	SF32	100	
1096	2	Output Active Power: Absolute Peak	-	W	SF32	100	
1098	2	Output Active Power: Daily Peak	-	W	SF32	100	
1100	2	Output Active Power: Feedback on Curtailment applied	-	W	SF32	100	
1102	2	Reactive Power: Feedback on Set Point applied	-	VAr	SF32	100	
1104	2	Power Factor: Feedback on Set Point applied	-	-	SF32	100	
1106	2	Mean Grid Frequency	-	Hz	SF32	100	
1108	2	Active Power on DC Input #1	-	W	SF32	100	
1110	2	Voltage on DC Input #1	-	V	SF32	100	
1112	2	Current on DC Input #1	-	A	SF32	100	
1114	2	Active Power on DC Input #2	-	W	SF32	100	TRIO-TM Only
1116	2	Voltage on DC Input #2	-	V	SF32	100	TRIO-TM Only
1118	2	Current on DC Input #2	-	A	SF32	100	TRIO-TM Only
1120	2	Internal Temperature	-	°C	SF32	100	DC Box Temperature The refresh time may be slightly higher than the other parameters
1122	2	Inverter Temperature	-	°C	SF32	100	DC/AC Converter Temperature
1124	2	Reserved					
1126	2	Isolation Resistance	-	MΩ	SF32	100	Start up Value
1128	2	Max Leakage Current	-	A	SF32	100	Max Leakage current in case of ground fault
1130	2	Active Power on DC Input #3	-	W	SF32	100	TRIO-TM Only
1132	2	Voltage on DC Input #3	-	V	SF32	100	TRIO-TM Only
1134	2	Current on DC Input #3	-	A	SF32	100	TRIO-TM Only
Modbus Register Map Version							
3650	1	Family product	71	-	ASCII String	-	Fixed value "G"
3651	1	Product model	116	-	ASCII String	-	Fixed value "t"
3652	1	Major release	51	-	ASCII String	-	Fixed value "3"
3653	1	Minor release	48	-	ASCII String	-	Fixed value "0"
3654	1	Build version	0	-	U16	-	



Note: Any Modbus request (Function #4) that starts and ends on available Modbus Data Address will not trigger any exception code, also if the request includes reserved register. Modbus request that starts or ends on reserved register will trigger an exception with code 2.

Any Modbus request on Ascii String or 32 bit data type (SF32 and U32) must include all the registers of the block otherwise the inverter will trigger an exception with code 2. For example any request for inverter serial number must include all the Modbus Data Address in the range 1009-1115.

Annex

Annex 1 Inverter Type

Integer Value	ASCII Value	Inverter Type
78	"N"	Photovoltaic Inverter Type
87	"W"	Wind Inverter Type

Annex 2 Country/Grid standard

Integer Value	ASCII Value	Country Standard
65	"A"	UL1741
66	"B"	Netherlands
67	"C"	Czech Republic
68	"D"	Canada
69	"E"	VDE 0126
70	"F"	France LL 2013
71	"G"	Greece
72	"H"	Hungary
73	"I"	ENEL Guida
74	"J"	CEI-016
75	"K"	AS-4777
76	"L"	Thailand PEA
77	"M"	BG C10-11 110%
78	"N"	Romania
79	"O"	Korea
80	"P"	Portugal
81	"Q"	China HV
82	"R"	Ireland
83	"S"	Spain RD 1699
84	"T"	Taiwan
87	"W"	BDEW
89	"Y"	Turkey HV
90	"Z"	Brazil
100	"d"	FranceLL 2014
101	"e"	VDE 4105

Integer Value	ASCII Value	Country Standard
103	“g”	JP 50Hz 400V
104	“h”	JP 60Hz 400V
106	“j”	CEI021 EX
107	“k”	Israel
108	“l”	Singapore
109	“m”	BG C10-11 100%
110	“n”	EN 50438
111	“o”	Corsica
112	“p”	Spain RD 1565
113	“q”	China LV
114	“r”	South Africa
115	“s”	Slovenia
116	“t”	Turkey LV
117	“u”	UK G59
119	“w”	VDE 0126 3W
121	“y”	Thailand MEA
88	“X”	DEBUG FF
120	“x”	DEBUG 88

Annex 3 Global State

Integer Value	Global State
0	Initializing (configuring power control)
1	Waiting sun or grid
2	Connecting to grid (checking grid)
3	Initializing (system startup)
4	Connecting to grid (switching-on DC/DC)
5	Connecting to grid (switching-on DC/AC)
6	Connected to grid
7	Post-alarm actions (recovery)
8	Post-alarm actions (pause)
9	Ground fault
10	Over-temperature fault
12	Connecting to grid (grid protection interface self-test)
13	Grid protection interface self-test fault
14	Connecting to grid (safety checks)

Integer Value	Global State
15	Leakage fault
24	Under-temperature fault
25	Interlock (remote off)
26	Interlock (emergency stop)
27	Executing auto-test
29	Grounding-kit fault
30	SW bundle not valid fault
41	Temperature sensors fault
42	Grid sequence fault
51	Arc fault
53	Arc detector self-test fault
116	Power stage off-line
118	Arc detector wrong configuration fault
119	Arc detector self-test
120	Configuration fault (bad model)
124	Latch with “Reset by hand”
150	Power stage communication fault
151	Configuration fault (bad global-settings)
200	Programming power stage



NOTE: the inverter can export power into the grid if and only if the global state of the inverter is Run (6)

Annex 4 Alarm State

Integer Value	Alarm State	Alarm Code
0	No alarm	NONE
2	Input OC	E001
3	Input UV	W002
4	Input OV	E002
6	No pars (DSP)	E003
7	Bulk OV	E004
8	Internal error	E005
9	Output OC	E006
10	IGBT sat.	E007
11	Bulk UV	W011

Integer Value	Alarm State	Alarm Code
14	Bulk UV	E010
15	Ramp Fault	E011
16	Internal error	E012
17	Wrong input mode (not available for TRIO-50/60-TL)	E013
19	Over temperature	E014
20	Cap. Fault	E015
21	Internal error	E016
22	Internal error	E017
23	Leak fault	E018
24	Internal error	E049
25	Internal error	E019
26	Internal error	E012
27	Internal error	E020
28	Internal error	E021
29	Internal error	E019
30	Internal error	E022
31	DC injection	E023
38	Riso low	E025
42	Mid bulk OV	E029
44	Internal error	E031
45	Internal error	E032
48	Undertemperature	E033
49	IGBT not ready	E034
50	Remote Off	E035
51	Internal error	E036
54	Riso low	E037
62	Island. Detected	W015
78	Riso Test Fail	E078
79	AFDD activated	E050
82	AFDD fault	E053
85	AFDD wrong conf.	E055
90	Periodic Grid Off	W048
91	Internal error	E077
95	Grid conn. fault	W046
96	Latch-Manual ent	E075
150	Update Incomplete	W047
152	Wrong Sequence	E079

Integer Value	Alarm State	Alarm Code
156	BackFeed OC	E084
167	Aux. Bulk Feed Conn. Error (not available for TRIO-50/60-TL)	E098
170	Ground Kit Self Test Error (not available for TRIO-50/60-TL)	E099



NOTE: The inverter may not be grid connected with a No Alarm State (0), to check if the inverter is grid connected refer to Global State Run condition (6)

Annex 5 DC/DC Converter State

Integer Value	DC/DC Converter State
1	Ramp
2	MPPT
4	Input over-current
5	Input under-voltage
6	Input over-voltage
7	Low input
8	No configuration
9	Bulk under-voltage
10	Communication error
11	Ramp fault
12	Pending redundancy fault
13	Wrong input mode
14	Ground fault
15	Pending redundancy fault
16	IGBT error
17	Leakage sensor self-test fault
18	Grid fault
19	Communication error
20	Charging bulk
21	IGBT not ready
255	Not programmed



NOTE: DC/DC State and DC/AC State does not include useful information for Monitoring Systems and are used only for Debug purpose. Refer to Global, Alarm and Derating States for Monitoring Systems development.

Annex 6 DC/AC converter state

Integer Value	DC/AC Converter State
0	Initializing
1	Connecting to grid
2	Connected to grid
3	Bulk over-voltage
4	Output over-current
5	IGBT fault
6	Bulk under-voltage
7	Leakage sensor degaussing error
8	No configuration
9	Low bulk voltage
10	Grid fault
11	Communication error
12	Leakage sensor degaussing error
13	Connecting to grid
14	Bulk capacitor fault
15	Leakage fault
16	Pending redundancy fault
17	Leakage sensor self-test fault
18	Grid protection self-test
19	Leakage sensor self-test
20	Grid protection self-test
21	Grid protection self-test
22	Grid protection self-test timeout
23	Grid protection self-test fault
24	Grid protection self-test fault
25	Auto-test
30	Grid voltage read error
31	Grid current read error
33	IGBT not ready
35	Communication error
36	Mid-bulk over-voltage
255	Not programmed



NOTE: DC/DC State and DC/AC State does not include useful information for monitoring systems and are used only for debug purpose. Refer to Global, Alarm and Derating dtates for monitoring systems development.

Annex 7 Derating State

LSB Bit Index	Derating State (TRUE = Derating active)
0	Power curtailment from user setpoint
1	Grid over-frequency derating
2	Average grid over-voltage derating
3	Anti-islanding derating
4	Nameplate grid current limitation
5	Over-temperature derating
6	DC over-voltage derating
7	Energy storage system zero-Power injection



NOTE: The Derating State is encoded as Bit Index and more than one derating may be active. In that case the output power will be limited to the lowest Derating among the active ones.

Example: A value 3 on the Derating State (Modbus Data Address=1054) corresponds to active deratings for

- Power Curtailment Set Point
- Grid over-frequency

A value 3 corresponds to the following Bit Index:

Bit 15	Bit 14	Bit 13	Bit 12	Bit 11	Bit 10	Bit 9	Bit 8	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Not Used								0	0	0	0	0	0	1	1

Annex 8 Transient options

The Transient options define the behavior of the power management during a transition between two different set points, the inverter shares the transient options for all the power control functions available: active power curtailment, reactive power and power factor set points.

The "transient" options include two different configuration modes; Transient Step on the Most Significant Byte (MSB) and the Smooth mode on the Less Significant Byte (LSB).

Each power management block, backward compatibility and lower plant control, manages the transient option independently, please set properly:

- Modbus Data Address 198 for block #1: Back compatibility configuration
- Modbus Data Address 505 for block #2: Power Plant Control.

Default settings

Modbus Data Address 505 or 198	
MSB - Transient Step	LSB - Smooth mode
0x00	0x00
Meaning: 4s base step	Meaning: Slope mode

Transient Step

The *Transient Step* defines the base time step considered on the LSB of the *Transient Time* and can be configure as follows:

Register Value	Byte position	Uint Value (with LSB=0)	Transient Step
0	0x00--	0	4s step
1	0x01--	256	1s step

For further details on Transient Time encoding check the examples on Annex 9

Smooth Mode

The *smooth mode* defines how the inverter manages a transition between two set points, the smooth mode can be configured as follows:

Register Value	Byte position	Uint Value (with MSB=0)	Smooth Mode
0	0x--00	0	Slope
1	0x--80	128	Time

Slope mode: If the slope mode is configured, the inverter will apply a fixed ramp during the transition between two different set points. In this case, the transient time (see annex 9) becomes the time necessary for the inverter to handle the power between the maximum and minimum set points manageable, according to the full operation range:

- Default Active power range $P_{range} = [0...50kW]$ or $[0...60kW]$
- Default Reactive power range $Q_{range} = [-50kVar...50kVar]$ or $[-60kVar...60kVar]$
- Default Power Factor range $\cos(\varphi)_{range} = [-1...1]$

Once the Transient Time (T), the range and the difference between previous and new set point (Δ) are fixed, the time required to reach the new set point and the slope will follow the formulas:

$$\text{Elapsed Time} = \begin{cases} \Delta P * \frac{T}{P_{range}} \\ \Delta Q * \frac{T}{Q_{range}} \\ \Delta \cos(\varphi) * \frac{T}{\cos(\varphi)_{range}} \end{cases} \quad \text{Slope} = \begin{cases} \frac{P_{range}}{T} \\ \frac{Q_{range}}{T} \\ \frac{\cos(\varphi)_{range}}{T} \end{cases}$$



Note: According to the formula the elapsed time change with the set points while the slope is fixed

Example: If we assume a transition on *Slope mode* between the following set points

- P1 = 10kW
- P2 = 30kW

with a Transient Time $T = 1.2s$, the elapsed time and slope will be the following:

$$\text{Elapsed Time} = (30kW - 10kW) * \frac{1.2s}{50kW} = 480 \text{ ms} \quad \text{Slope} = \frac{50kW}{1.2s} = 41,67 \text{ kW/s}$$

While in case of a transition between the following set points

- P1 = 10kW
- P2 = 50k

With a Transient Time $T = 1.2s$, the elapsed time and slope will be the following:

$$\text{Elapsed Time} = (50kW - 10kW) * \frac{1.2s}{50kW} = 960 \text{ ms} \quad \text{Slope} = \frac{50kW}{1.2s} = 41,67 \text{ kW/s}$$

Time mode: If the time mode is configured, the inverter will execute the transition between two different set points within a fixed time, the Transient Time (see annex 9) becomes the time elapsed to reach the new set point.

Once the Transient Time (T) and the difference between previous and new set point (Δ) is fixed, the time required to reach the new set point and the slope will follow the formulas:

$$\text{Elapsed Time} = T \quad \text{Slope} = \frac{\Delta P}{T} \text{ or } \frac{\Delta Q}{T} \text{ or } \frac{\Delta \cos(\varphi)}{T}$$



Note: According to the formula the elapsed time is fixed while the slope change with the set points

Example: If we assume a transition on *Time mode* between the following set points

- P1 = 10kW
- P2 = 30kW

With a Transient Time $T = 1.2s$, the elapsed time and slope will be the following:

$$\text{Elapsed Time} = 1200 \text{ ms} \quad \text{Slope} = \frac{(30kW - 10kW)}{1.2s} = 16,67 \text{ kW/s}$$

While in case of a transition between the following set points

- P1 = 10kW
- P2 = 50k

With a Transient Time $T = 1.2s$, the elapsed time and slope will be the following:

$$\text{Elapsed Time} = 1200 \text{ ms} \quad \text{Slope} = \frac{(50kW - 10kW)}{1.2s} = 33,34 \text{ kW/s}$$

Transient Step summary table

The Transient Options can assume the following values:

Transient Options value (Unit)	Modbus register value (Hex)	Transient Step	Smooth mode
0	0x0000	4s step	Slope
128	0x0080	4s step	Time
256	0x0100	1s step	Slope
384	0x0180	1s step	Time



NOTE: The values on table are the only ones that should be written on the Transient Options register.

Annex 9 Transient Time

The transient time defines the time elapsed to reach the set point when a new power command is received by the inverter. The value and meaning of Transient Time depend on the configuration of the Transient Options register

The transient time is calculated as follows:

$$T = N * S + \frac{M}{100} \text{ seconds}$$

Where S is the Transient Step defined by Transient Options register (see annex 8), N is the less significant Byte (LSB) of the Transient Time register expressed as multiple of S and M is the most significant byte (MSB) of the Transient Time register expressed as multiple of 10ms.

To easily evaluate the *Transient Time* T it is possible to use a simplified formula, if we consider $INT(T)$ as the integer part and $DEC(T)$ as the decimal part of the *Transient Time* that we want to apply to the inverter, then it is necessary to write the Modbus register as follows:

- For Transient Step =1s :
Modbus Data Address 503 or 190 = $N + \frac{M}{100} \text{ seconds} = INT(T) + DEC(T) * 100 * 256$
- For Transient Step =4s:
Modbus Data Address 503 or 190 = $4 * N + \frac{M}{100} \text{ seconds} = INT(T/4) + DEC(T) * 100 * 256$

The *Transient Time* T is expressed on seconds



NOTE: In case of *Transient Step* = 4s it is possible to set the exact *Transient Time* only if $INT(T)$ is a multiple of 4.

Example: If we assume a *Transient Options* register set to 256 (1s base step with slope mode), then to set a *Transient Time* of 2.4s, it is necessary to set the Modbus Data Address 503 as follow:

- $INT(T) = 2;$
- $DEC(T) = 0.4$

Modbus Data Address 503/190 = $2 + 0.4 * 100 * 256 = 10242$ (0x2802)

The inverter will apply the new active power set point with a fixed slope of 20.83 kW/s

Example: If we assume a *Transient Options* register set to 128 (4s base step with Time mode), to set a *Transient Time* of 12.2s, it is necessary to set the Modbus Data Address 503 as follow:

- $INT\left(\frac{T}{4}\right) = \frac{12}{4} = 3;$
- $DEC(T) = 0.2$

Modbus Data Address 503/190 = $3 + 20 * 256 = 5123$ (0x1403)

The inverter will apply the new active power set point in 12.2 seconds.

Example: If we assume a *Transient Options* register set to 0 (default value with 4s base step and slope mode) and the Modbus register 503 set to 12805 (0x3205), then the Transient Time is 20.5 seconds and the inverter will apply the new active power set point with a fixed slope of 2.4 kW/s.

Value =1330	MSB	LSB
Hex	0x32	0x05
Integer	50	5
$T = 4 * 5 + \frac{50}{100} = 20.5 \text{ seconds}$		

Annex 12 Exception code description

Code	Name	Meaning
1	Illegal Function	The function code received in the query is not an allowable action for the slave. This may happen when the function code is only applicable or is not implemented in the unit selected. It could also indicate that the slave is in the wrong state to process a request of this type, for example because it is not configured and is being asked to return register values.
2	Illegal Data Address	The data address received in the query is not an allowable address for the slave. More specifically, the combination of starting address and length is invalid.
3	Illegal Data Value	A value contained in the query data field is not an allowable value for the slave. This indicates a fault in the structure of the remainder of a complex request, such as that the implied length is incorrect.
4	Server Device Failure (Server = Slave)	A value contained in the query data field is not an allowable value for the slave. This indicates a fault in the structure of the remainder of a complex request, such as that the implied length is incorrect.
6	Server Device Busy (Server = Slave)	Specialized use in conjunction with programming commands. The slave is engaged in processing a long-duration program command. The master should retransmit the message later when the slave is free.

Annex 13 Modbus CRC coding example

Example of procedure for generating a Modbus CRC:

1. Load a 16-bit register with FFFF hex (all 1's). Call this the CRC register.
2. Exclusive OR the first 8-bit byte of the message with the low-order byte of the 16-bit CRC register, putting the result in the CRC register.
3. Shift the CRC register one bit to the right (toward the LSB), zero-filling the MSB. Extract and examine the LSB.
4. (If the LSB was 0): Repeat Step 3 (another shift).
(If the LSB was 1): Exclusive OR between the CRC register and the polynomial value 0xA001 (1010 0000 0000 0001).
5. Repeat steps 3 and 4 until 8 shifts have been performed. When this is done, a complete 8-bit byte will have been processed.
6. Repeat steps 2 through 5 for the next 8-bit byte of the message. Continue to operate in this way until all bytes have been processed.
7. The final content of the CRC register is the CRC value.
8. When the CRC is placed into the message, its upper and lower bytes must be swapped.

Example of CRC calculation (frame 02 07)

CRC register initialization		1111	1111	1111	1111
XOR 1st character		0000	0000	0000	0010
		1111	1111	1111	1101
	Move 1	0111	1111	1111	1110 1
Flag to 1, XOR polynomial		1010	0000	0000	0001
		1101	1111	1111	1111
	Move 2	0110	1111	1111	1111 1
Flag to 1, XOR polynomial		1010	0000	0000	0001
		1100	1111	1111	1110
	Move 3	0110	0111	1111	1111 0
	Move 4	0011	0011	1111	1111 1
		1010	0000	0000	0001
		1001	0011	1111	1110
	Move 5	0100	1001	1111	1111 0
	Move 6	0010	0100	1111	1111 1
		1010	0000	0000	0001
		1000	0100	1111	1110
	Move 7	0100	0010	0111	1111 0
	Move 8	0010	0001	0011	1111 1
		1010	0000	0000	0001
		1000	0001	0011	1110
XOR 2nd character		0000	0000	0000	0111
		1000	0001	0011	1001
	Move 1	0100	0000	1001	1100 1
		1010	0000	0000	0001
		1110	0000	1001	1101
	Move 2	0111	0000	0100	1110 1
		1010	0000	0000	0001
		1101	0000	0100	1111
	Move 3	0110	1000	0010	0111 1
		1010	0000	0000	0001
		1100	1000	0010	0110
	Move 4	0110	0100	0001	0011 0
	Move 5	0011	0010	0000	1001 1
		1010	0000	0000	0001
		1001	0010	0000	1000
	Move 6	0100	1001	0000	0100 0
	Move 7	0010	0100	1000	0010 0
	Move 8	0001	0010	0100	0001 0

Most significant
least significant

The CRC 16 of the frame is then: 4112

Annex 14 Model Encoding

The inverter model is an Ascii encoded string, refer to the following table to retrieve the association between inverter model and data value.

Register Integer Value	Register Exadecimal Value	Register Ascii Coding	Inverter Model
40	28	"("	TRIO-50-TL-OUTD-EU
47	2F	"/"	TRIO-60-TL-OUTD
41	29	")"	TRIO-50-TL-OUTD-US
58	3A	"."	TRIO-60-TL-OUTD-US
59	3B	"."	TRIO-50-TL-OUTD-JP
92	5C	"\"	TRIO-TM-50.0-400 TRIO-TM-60.0-480

Document revisions

Revision	Date	Modbus Map Version	ChangeLog
Rev 1.0	09/11/2016	GU00.0	Document created
Rev 2.0	20/03/2017	Gt03.0	Note added respect to the previous map: ▪ Chapter Capability ▪ Set points available and figure 3 ▪ Reactive power priority ▪ behavior in the reading of holding registers (Power Management) ▪ Night time reactive power ▪ DC Box Temperature register Modbus data address added respect to the previous map: ▪ Holding 182, 801, 802 for “Reset by Hand” feature ▪ Input (MPPT) Corrections made respect to the previous document release ▪ Changed unit (Wh) on energy measures Important notes: ▪ Since this release the reactive power on modbus data address 220 and 222 is managed through $\cos\phi$ instead of $\sin\phi$
Rev 2.1	19/05/2017	Gt03.0	Corrections made respect to the previous document release ▪ Updated the Annex 4 table
Rev 2.2	19/06/2017	Gt03.0	Note added respect to the previous map: ▪ Compatibility with inverter family TRIO-60. ▪ Annex 14 .
Rev 3.0	31/01/2018	Gt03.0	Note added respect to the previous map: ▪ Modbus map for inverter family TRIO-TM. Modbus data address added respect to the previous map: ▪ Registers for MPPT#2 and MPP #3
Rev 3.1	31/01/2018	Gt03.0	Corrections made respect to the previous document release ▪ Changed model designation
Rev 3.4	10/06/2019	Gt03.0	Corrections made respect to the previous document release ▪ Updated the Annex 4 table ▪ Removed the section Default settings

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